

Use proportional reasoning in scale, rates, and financial contexts

Look carefully. Think about parts of a whole.
Show your work by colouring, drawing, or writing.

Step 11

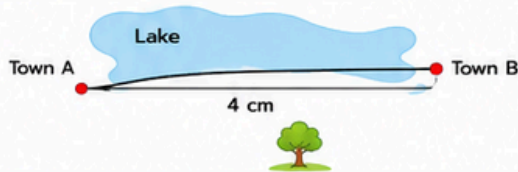
Early Elementary

★ Today I am learning to use proportional reasoning in scale, rates, and financial contexts.

1. Map Scale

The map scale is 1 cm = 50 km.

Look at the map and complete the table.



Journey	Map Distance (on map)	Real Distance (in km)
Town A to Town B	4 cm	_____ km
Town A to Lake	2 cm	_____ km
Lake to Town B	3 cm	_____ km
Town A to Park	1.5 cm	_____ km

2. Speed and Time (Rates)

A car travels at a constant speed of 60 km per hour.

Use the rate to complete the table.



Time (hours)	Distance (km)	How far?
1	60	60 km
2	_____	_____ km
3	_____	_____ km
4	_____	_____ km
5	_____	_____ km
$\frac{1}{2}$ (half)	_____	_____ km
$1\frac{1}{2}$ (one and a half)	_____	_____ km

3. Unit Rate

Apples cost \$3 for 6 apples.

Find the unit rate and complete the table.



Number of Apples	Total Cost	Cost per Apple
6	\$3	\$ _____
12	\$ _____	\$ _____
18	\$ _____	\$ _____
24	\$ _____	\$ _____
30	\$ _____	\$ _____

Unit rate means the cost of 1 apple.

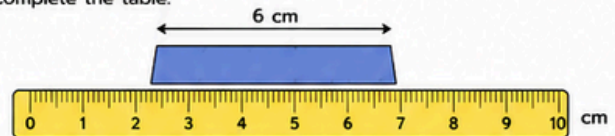


4. Scale Drawing

The scale of the drawing is 1 cm = 2 m.

Use the scale to find the real length.

Complete the table.



Drawing Length (on the picture)	Scale	Real Length (in metres)
6 cm	1 cm = 2 m	_____ m
8 cm	1 cm = 2 m	_____ m
12 cm	1 cm = 2 m	_____ m
2.5 cm	1 cm = 2 m	_____ m

5. Recipe (Proportional Quantities)

A recipe uses 2 cups of flour to make 12 muffins.
Use proportional reasoning to complete the table.



Number of Muffins	Cups of Flour	Work it Out
12	2	_____
24	_____	_____
36	_____	_____
6	_____	_____
18	_____	_____

6. Money and Discounts

A shirt costs \$40.

Complete the table.



Discount	Percent Off	Amount Off	Sale Price
10% off	10%	\$ _____	\$ _____
25% off	25%	\$ _____	\$ _____
50% off	50%	\$ _____	\$ _____

7. Real-Life Problems

Solve each problem.

- a) A train travels 120 km in 2 hours at a constant speed. How far will it travel in 5 hours at the same speed?



Answer: _____ km

- b) 4 notebooks cost \$6. How much will 10 notebooks cost at the same price?



Answer: \$ _____

- c) A map uses the scale 1 cm = 25 km. Two towns are 6.5 cm apart on the map. What is the real distance between the towns?



Answer: _____ km

8. Think and Answer

- a) What does proportional reasoning mean? _____
- b) Give one example of proportional reasoning in real life. _____
- c) Why is it important to use a scale or a rate correctly? _____

Look for the relationship and use it to solve problems!

I can...



- ☐ use scale to find real lengths and distances.
- ☐ use rates to find distance, time, and unit rates.
- ☐ solve problems involving proportional reasoning.
- ☐ use proportional reasoning in money and discounts.



My Learning Goal

I will use proportional reasoning in different situations.

